

20-Mode Spatial and Polarization Multiplexer using Vertically Stacked Three-Layer Dielectric Metasurfaces

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Abstract A spatial and polarization multiplexer based on compact stacked metasurfaces is experimentally demonstrated. Using three metasurface layers with 0.8-mm-diameter aperture, we achieve simultaneous conversion of 20 input beams to the desired Hermite–Gaussian modes at 1550 nm, demonstrating a scalable optical interface for space-division-multiplexed systems. ©2026 The Author(s)

Introduction

The continued growth of optical communication systems is driving demand for higher transmission capacity and bandwidth density, motivating the development of space-division multiplexing (SDM) and high-density optical interconnect technologies [1,2]. These applications require compact and scalable interfaces for generating, separating, and rearranging multiple spatial channels. Existing technologies, including fiber-based fan-in/fan-out (FIFO) devices, photonic lanterns, 3D waveguide devices, and multi-plane light conversion (MPLC), have demonstrated efficient spatial-channel interfacing and mode manipulation [3,4]. Yet, they still leave room for improvement in compactness and integration, as well as in enabling simultaneous spatial and polarization multiplexing in a single device.

Dielectric metasurfaces (MSs) have recently attracted attention as substantially more compact multifunctional optical interfaces for communication systems, including coherent receivers, mode multiplexers, and FIFO devices [5–7]. To extend these capabilities to higher mode counts and simultaneous control of spatial and polarization modes, multiple birefringent MS layers are

generally required. Based on this concept, a folded multi-layer MS device for 6-mode vectorial mode conversion and a double-sided MS device for FIFO coupling of a 4-core multicore fiber have been demonstrated [8,9]. However, the folded configuration relies on oblique input/output access, restricting straightforward integration with other optical or electronic components. The double-sided configuration, meanwhile, is inherently limited to two MS layers.

In this work, we experimentally demonstrate, to the best of our knowledge, the first vertically stacked three-layer dielectric MS device for spatial and polarization mode multiplexing of normally incident beams. The device simultaneously converts 20 spatially separated single-mode beams arranged in a 2D array into 20 orthogonal modes consisting of 10 Hermite–Gaussian (HG) spatial modes in each of two orthogonal polarizations. Simulations predict 20-mode operation with an average mode-matching penalty of 0.05 dB. A proof-of-concept device is fabricated on a silicon-on-quartz (SOQ) substrate, exhibiting average mode-matching penalties of 0.9 dB and 2.4 dB for X and Y polarizations, respectively. These results demonstrate the potential of vertically stacked

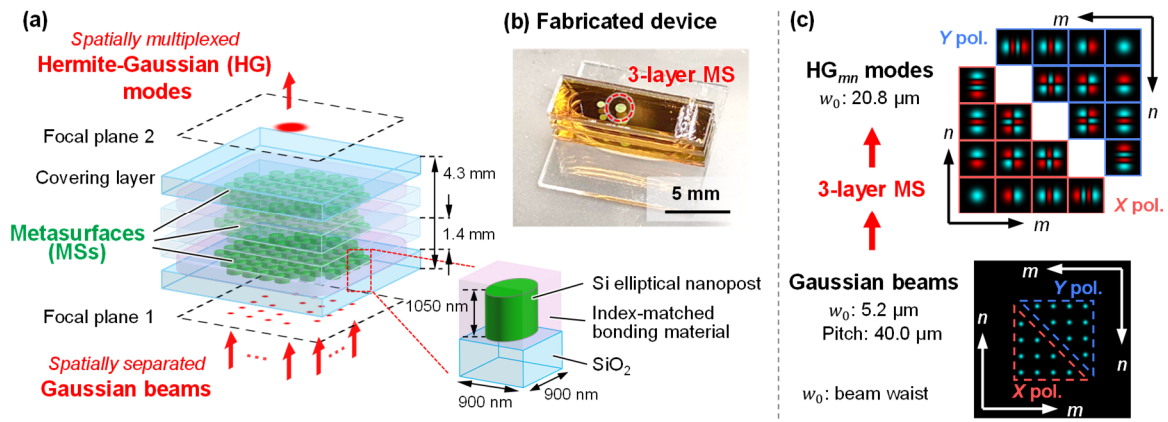


Fig. 1: Overview of the demonstrated stacked metasurface (MS)-based spatial and polarization mode multiplexer.

- (a) Three-dimensional view of the device structure, with an inset showing the birefringent elliptical nanopost.
(b) Optical microscope image of the fabricated device. (c) Functionality of the device, showing the definitions of the input Gaussian beams and output Hermite-Gaussian (HG) modes in the two orthogonal polarizations.

transmissive MSs as a compact and scalable platform for spatial and polarization mode manipulation in SDM and dense optical interconnect systems.

Device concept and operating principle

Fig. 1 provides an overview of the demonstrated 20-mode (10 spatial modes $\times$ 2 polarization modes) multiplexer. Figs. 1(a) and 1(b) show the device structure and an optical microscope image of the fabricated device, respectively. The device comprises three vertically stacked MSs, each formed on a 625- μm -thick SiO_2 substrate with a glass cover layer. Each MS layer has a diameter of 800 μm . The inset of Fig. 1(a) shows the birefringent meta-atom employed in this work, which consists of an elliptical Si nanopost. These nanoposts are 1050 nm in height and arranged on a 900-nm lattice, enabling local polarization-dependent control of the transmitted optical field. The three MSs are stacked vertically via an index-matched ultraviolet (UV) cured bonding material with an interlayer spacing of 1.4 μm . As the light incident normal to the device surface propagates through the three MS layers, desired mode conversion is achieved.

Fig. 1(c) explains the mode-multiplexing functionality of the device. Each MS layer is optimized for operation at a wavelength of 1550 nm to

convert 10 X-polarized and 10 Y-polarized Gaussian beams arranged on a 40- μm -pitch array at focal plane 1 into 20 HG modes in the same lateral position at focal plane 2. The output consists of 10 HG modes in each of the two orthogonal polarizations, spatially multiplexed while preserving the input polarization. The beam waists w_0 of the input Gaussian beams and output HG modes are set to 5.2 μm and 20.8 μm , respectively.

Numerical design and simulated performance

We employed the adjoint-optimization-based method [8] to optimize the geometries of the nanoposts in all three MS layers simultaneously to maximize the mode conversion fidelity while suppressing inter-mode crosstalk (XT) and mode-dependent loss (MDL).

We first confirmed that at least three MS layers were required to achieve an average coupling loss below 1 dB for the 20-mode device. We then optimized the MS structure numerically and evaluated the resulting device performance. Fig. 2 shows the designed phase shift distributions for the X- and Y-polarized components in each of the three MS layers. Based on these results, we then determined the radii of the elliptical nanoposts along the X and Y directions to realize the required phase shifts for the two polarizations.

Fig. 3 shows the calculated coupling matrix $\mathbf{C}$, whose elements are defined as

$$C_{ij} = \left| \iint dx dy \mathbf{E}_i^{\text{tar}}(x, y)^\dagger \mathbf{E}_j^{\text{out}}(x, y) \right|^2, \quad (1)$$

where $\mathbf{E}_i^{\text{tar}}(x, y)$ is the target vectorial field of the i -th HG mode and $\mathbf{E}_j^{\text{out}}(x, y)$ is the output vectorial field obtained for the j -th input mode. The calculated performance indicates an average coupling loss of 0.9 dB, XT below -23 dB, and MDL of 0.5 dB. Separately, the average mode-matching penalty, defined from the normalized overlap between the obtained and target fields, is 0.05 dB.

Device fabrication and characterization

We then fabricated each MS layer using a procedure based on that described in [10,11], followed by a stacking process. The stacking accuracy was estimated to be 2 μm for the lateral alignment and 5 μm for the interlayer spacing. Fig. 4 shows an optical micrograph and an SEM image of the fabricated 3rd MS layer on an SOQ substrate before stacking with other layers.

The fabricated device was characterized using the digital off-axis holography technique [12], with the setup shown in Fig. 5(a). For reconstruction of the complex optical field distribution, the light transmitted through the device was interfered with a reference beam and captured by a near-infrared (NIR) camera. A two-dimensional

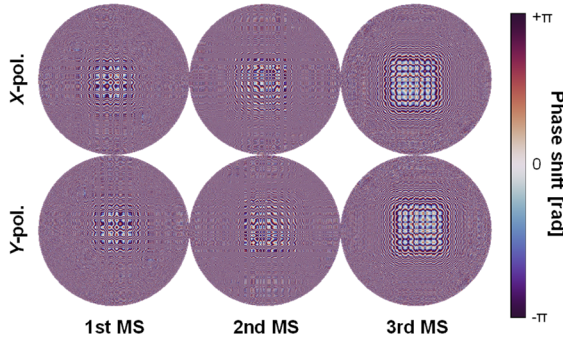


Fig. 2: Designed phase shift distributions in the three MS layers for X and Y polarizations.

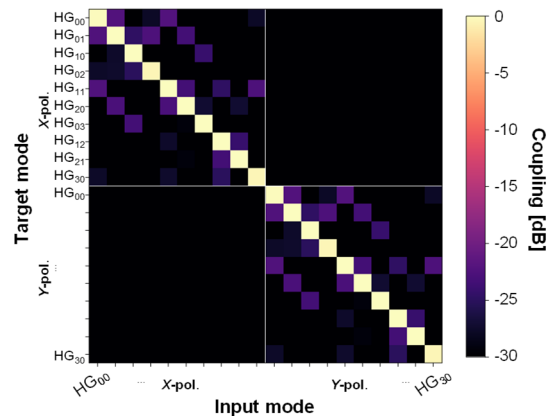


Fig. 3: Calculated coupling matrix $\mathbf{C}$ for the multiplexer.

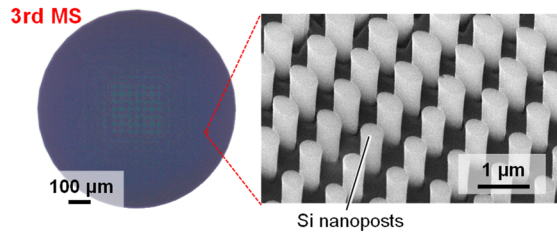


Fig. 4: Optical micrograph and SEM image of the fabricated 3rd MS layer on a silicon-on-quartz substrate.

galvanometric mirror was used to scan the input beam position.

The reconstructed complex fields of the output modes for X and Y polarizations at a wavelength of 1550 nm are shown in Fig. 5(b). We confirm that each input mode is converted to the corresponding HG mode with the intended polarization. The Y-polarized outputs exhibit larger distortions, which are mainly attributed to interlayer misalignment and residual tilt along the Y direction. Fig. 5(c) shows the measured coupling matrix C , which is obtained from the overlap integrals between the reconstructed output fields and the target HG modes. The maximum XTs estimated from C are -13.5 dB and -8.4 dB for X and Y polarizations, respectively, whereas the mode-matching penalties are 0.9 dB and 2.4 dB. The measured average insertion loss is 8 dB. The observed degradation is attributed to a combination of device imperfections, including fabrication errors, interlayer misalignment, and layer tilt, as well as possible asymmetry in the measurement setup. It should therefore be mitigated by improving the MS fabrication and characterization procedures.

Conclusions

We have proposed and demonstrated a vertically stacked three-layer dielectric MS device for spatial and polarization mode multiplexing. The fabricated device successfully converted a 2D array of input Gaussian beams into spatially and polarization multiplexed 20 orthogonal HG modes, exhibiting average mode-matching penalties of 0.9 dB and 2.4 dB for X and Y polarizations, respectively. The entire device has 0.8-mm aperture and 4.3-mm thickness and is compatible with direct facet attachment to multi-mode fibers and photonic integrated circuits. The presented results demonstrate the potential of vertically stacked transmissive MSs as a compact and scalable platform for high-dimensional spatial and polarization mode manipulation in SDM and dense optical interconnect systems.

Acknowledgements

This work was obtained in part from the commissioned research (No. JPJ012368C08801) by NICT, Japan, and JST-ALCA-Next Japan (No. JPMJAN25F1). The SOQ wafer was provided by Shin-Etsu Chemical Co., Ltd. The device was fabricated in part using cleanroom facilities of d.lab at the University of Tokyo, supported by MEXT ARIM Program (No. JPMXP1225UT1102).

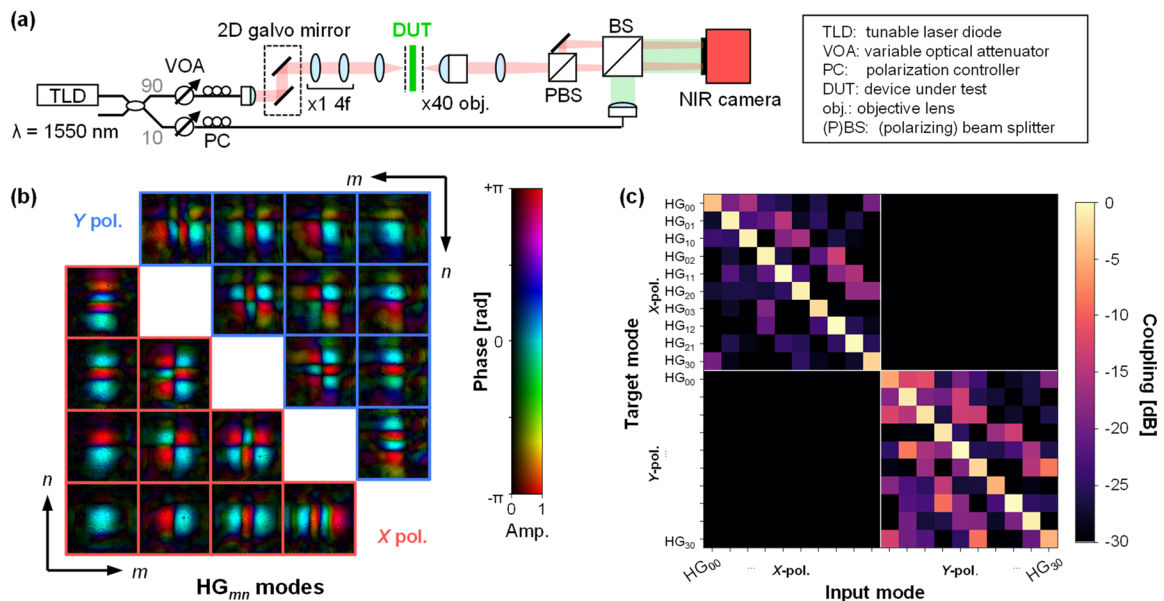


Fig. 5: Experimental characterization of the fabricated device.

(a) Experimental setup. (b) Reconstructed complex fields of the output modes. (c) Measured coupling matrix C .

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